Phase 17: User Acceptance Testing (UAT) Plan

Document Code: UAT-PLN-001

Project: Duplicate Bin Ecosystem

Target Release: v0.2.0 Desktop + v1.0.0 SaaS Hybrid

Lead Author: Senior Technical Business Analyst / QA Analyst Lead

1. The Role of the Business Analyst in UAT

In enterprise software delivery, the Business Analyst serves as the **custodian of business intent** during User Acceptance Testing. While Quality Assurance (QA) engineers test against technical specifications and system boundaries, the BA ensures the system solves the original business problem and satisfies real-world user workflows:

1. **UAT Test Scenario Authoring:** Translating business requirements (BRD/FRD) and user stories into realistic business scenarios.
2. **Business User Facilitation:** Guiding end-user stakeholders (photographers, IT admins, agency owners) through test scripts without technical jargon.
3. **Defect Triage & Impact Classification:** Determining whether an unexpected system behavior constitutes a critical business blocker, a minor defect, or an out-of-scope change request.
4. **Traceability Governance:** Ensuring all business acceptance criteria defined in the Requirements Catalog are explicitly exercised and signed off.
5. **Formal Acceptance Recommendation:** Delivering the final UAT Summary Report and recommending Go/No-Go release decisions to the Product Owner and Executive Sponsor.

2. UAT Scope & Boundaries

In-Scope for UAT

- End-to-end directory scanning across mixed-media repositories (photos, videos, PDFs, code).
- Validation of two-pass size pre-filtering and cryptographic hash accuracy.
- Execution of all 8 automated selection heuristics (`Keep Best Quality`, `Keep Newest`, etc.).

- Non-destructive archiving to quarantine and 1-click restore verification (`Ctrl+Z`).
- Workstation pairing via 6-digit codes and remote scan triggering.
- Credit deduction ledger accuracy upon scan completion.
- Usability of drag-and-drop ingestion and keyboard shortcuts.

Out-of-Scope for UAT

- Code-level unit testing and performance benchmarking (handled in internal QA sprint cycles).
- Direct cloud API crawling without local folder synchronization.
- Operating system stress testing under kernel failure conditions.

3. Test Environment & Prerequisites

Environment Component	Specification
Client OS Workstations	Windows 11 Pro (64-bit), Ubuntu 22.04 LTS, macOS Sonoma (M-series / x86_64).
SaaS Cloud Backend	Local or staging Uvicorn/FastAPI server connected to SQLite/PostgreSQL staging instance.
Test Data Repository	Synthetic and real-world directory structures containing: • 1,000 files with known duplicates. • 10,000 files with mixed extensions (RAW, JPEG, MP4, PDF). • Zero-byte files, symlinks, and read-only files.
Network Configuration	Active local network connection for WebSocket testing; offline mode validation toggle.

4. UAT Entry & Exit Criteria

```
┌──────────────────────────────────────────────────────────────────────┐
│ UAT ENTRY CRITERIA (Must be 100% satisfied before UAT begins)        │
├──────────────────────────────────────────────────────────────────────┤
│ 1. 100% of unit and integration tests passing in CI/CD pipeline.     │
│ 2. Zero Severity-1 (Blocker) or Severity-2 (Critical) defects open.  │
```

```
3. Staging backend deployed and verified via healthcheck endpoints.
4. Test datasets and seed directories staged across target workstations.
5. UAT test scripts approved by Business Analyst and Product Owner.
```

```
UAT EXIT CRITERIA (Must be satisfied for Go/No-Go release sign-off)
1. 100% of Must-Have (P1) UAT test cases executed and passed.
2. 0 open critical business-blocking defects.
3. 100% data recovery verified during quarantine rollback testing.
4. Business Owner and IT Admin stakeholders have formally signed off.
5. User documentation and quick-start guides match actual software behavior.
```

5. Defect Severity & Triage Protocol

Severity Level	Definition	Business Impact	SLA for Resolution
Sev-1 (Blocker)	Permanent data loss, system crash, or complete scan freeze.	Halts all business operations; unacceptable release risk.	4 Hours (Immediate hotfix)
Sev-2 (Critical)	Core feature failure (e.g., selection strategy selects all files; pairing fails).	Major workflow blocked; no viable business workaround.	24 Hours
Sev-3 (Major)	Secondary feature failure (e.g., image thumbnail fails to render; shortcut fails).	Annoying to user, but viable manual workaround exists.	Within Current Sprint
Sev-4 (Minor)	Minor UI alignment, cosmetic dark-theme discrepancy, or typo.	Trivial cosmetic impact; does not hinder functionality.	Backlog Prioritization